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1 /Users/anuragpaul/Library/Java/JavaVirtualMachines/
  corretto-20.0.2.1/Contents/Home/bin/java -Dvaadin.
  copilot.pluginDotFilePath=/Users/anuragpaul/Data-
  Structre-Java/.idea/.copilot-plugin -javaagent:/
  Applications/IntelliJ IDEA CE.app/Contents/lib/
  idea_rt.jar=58606 -Dfile.encoding=UTF-8 -Dsun.
  stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -
  classpath /Users/anuragpaul/Data-Structre-Java/
  target/classes org.example.assignment5.Main
2 DFS Performance:
3   n=100   E=236   time=261834 ns
4   n=500   E=6167  time=1540500 ns
5   n=1000  E=24894  time=5394750 ns
6
7 BFS Performance:
8   n=100   E=236   time=295459 ns
9   n=500   E=6167  time=902916 ns
10  n=1000  E=24894  time=2934375 ns
11
12 Dijkstra Performance:
13  n=100   E=236   time=1461125 ns
14  n=500   E=6167  time=2311792 ns
15  n=1000  E=24894  time=7253917 ns
16
17 Graph Demo:
18 Vertices: 6   Edges: 9
19
20 DFS from A:
21   Visit order: [A, C, E, F, D, B]
22   Discovery tree edges:
23     A --> B   (weight 4.0)
24     B --> C   (weight 1.0)
25     C --> D   (weight 8.0)
26     D --> E   (weight 2.0)
27     E --> F   (weight 3.0)
28
29 BFS from A:
30   Visit order: [A, B, C, D, E, F]
31   Hop distances from A:
32     A : 0
33     B : 1
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34      C : 1
35      D : 2
36      E : 2
37      F : 3
38
39 Dijkstra Shortest Paths from A:
40   A -> A : 0.0   path: [A]
41   A -> B : 3.0   path: [A, C, B]
42   A -> C : 2.0   path: [A, C]
43   A -> D : 8.0   path: [A, C, B, D]
44   A -> E : 10.0  path: [A, C, B, D, E]
45   A -> F : 13.0  path: [A, C, B, D, E, F]
46
47 Shortest path A --> F: [A, C, B, D, E, F]
48 Total distance: 13.0
49
50 Process finished with exit code 0
51
```